

CHAPTER 15

PERFORMANCE PREPARATION, MOBILITY, AND FLEXIBILITY

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After completing this chapter, you will be able to

- identify the components and benefits of a preperformance preparation routine;
- structure effective preperformance preparation routines (warm-ups);
- identify the factors that affect flexibility;
- use flexibility exercises that take advantage of proprioceptive neuromuscular facilitation; and
- select and apply appropriate static and dynamic stretching methods.

This chapter is devoted to two key areas: performance preparation (warming up) and training for flexibility. While these two areas often have been linked, it is important to differentiate between the two, because they have distinctly different key functions. A warm-up is designed to prepare an athlete for performance either in an upcoming training session or competition and where the goal is to

optimize performance while potentially lessening the risk of injury. Flexibility training, on the other hand, aims to increase the range of motion around a joint, normally through the use of different forms of stretching. The first part of this chapter looks at the aims of a warm-up and suggests appropriate structures and protocols for designing effective warm-ups, and the second part discusses factors that affect flexibility and the use of different stretching protocols to enhance flexibility.

Performance Preparation—The Warm-Up

A warm-up period is now almost universally accepted as an integral part of any training session or competition (10), yet until recently there was surprisingly little published research to support the practice (65) and even less on the optimal structure and content of a warm-up (48). Essentially, the overarching goal of a warm-up is to prepare an athlete mentally and physically for exercise or competition (45). A period of preparation is needed because a body at rest is unprepared for optimal performance. A period of warming up facilitates the instigation of a range of acute physiological responses that enable the body to cope with the specific demands being placed on it (65). Taken together, these acute responses not only enable the body to cope with these acute demands but also facilitate responses that can potentially optimize subsequent performance. While these processes occur across a number of body systems, they typically can be divided into temperature-related effects and non-temperature-related effects (10).

Temperature-related effects include an increase in muscle temperature, core temperature (62), enhanced neural function, and the disruption of transient connective tissue bonds (30), while non-temperature-related effects can include increased blood flow to muscles, an elevation of baseline oxygen consumption, and postactivation potentiation (10). While passive warm-up techniques (e.g., external heating of the muscles) may be able to generate some of the temperature-related effects, active warm-ups (using exercises or activity to elicit the required response) are generally preferred and advised because they are able to stimulate both the temperature- and non-temperature-related effects and can also affect the metabolic pathways associated with the upcoming activity (65). While ultimately the exact acute responses will depend on the

activity undertaken, the positive effects on performance are thought to include the following:

- Enhanced muscle fiber contraction frequency, conduction, and relaxation velocity of both agonist and antagonist muscles (45)
- Improvements in the rate of force development and reaction time (3)
- Improvements in muscle force output, especially at higher velocities (30)
- Lowered viscous resistance in muscles and joints (30)
- Improved oxygen delivery due to the Bohr effect, whereby higher temperatures facilitate oxygen release from hemoglobin and myoglobin (see [chapter 6](#) for more details) (62)
- Increased blood flow to active muscles through vasodilation (62)
- Enhanced metabolic reactions (30) facilitating energy transduction including faster ATP turnover, especially in anaerobic pathways (4)
- An increased psychological preparedness for performance (10)

While the number of quality studies investigating the impact of warm-up on performance is surprisingly low, more than 80% of the published research has demonstrated a positive effect on performance (37, 40). These performance improvements include enhancement of endurance performance (both aerobic and anaerobic), improvements in performance on physical tasks such as jumping, and in actual sporting performance (37).

Whenever an analysis of the effects of warm-up is carried out, or where recommendations as to the optimal form of warm-up are required, it is crucial to consider that the direct effects will be related to the structure and content of the warm-up, the characteristics of the athletes concerned, and the nature of the subsequent activity (37). As a result, while general findings can be identified and recommendations made, a degree of specificity is required in any scenario. Consequently, it is highly unlikely that a perfect warm-up

exists, and in all probability, a range of warm-ups can be used, provided that they are structured in a way that addresses the specific physiological, biomechanical, and psychological requirements of the upcoming activity, take into consideration the characteristics of the athletes concerned, and use an approach that exploits the specific acute responses to activity that can influence performance.

Given that the structure and content of the warm-up influences potential responses, the warm-up needs to be planned in relation to the activity to be performed, the physiological responses required, the characteristics of the athletes in question, and the logistical requirements of the situation.

In addition to a focus on acute performance, effective warm-ups traditionally have been thought to decrease the risk of injury. While the influence of a warm-up on injury prevention is unclear, the evidence suggests that positive effects may exist (36, 84, 85). For example, increased muscle temperature can increase resistance to muscle tears (78).

Components of a Warm-Up

While the precise nature of the warm-up will be closely related to context, an effective warm-up generally follows these processes and is built progressively (46):

- A stimulation of the cardiorespiratory and metabolic processes that facilitate energy transduction
- A progressive raising of body and working muscle temperatures
- An increase in the range of motion in relation to that required in the upcoming activity
- A progressive increase in muscle contraction intensity relative to the activities in the upcoming session or competition

As stated previously, a body at rest is unprepared for activity, and in this way a warm-up can be considered a process of preparation for the upcoming performance. The conventional advice is that a warm-up should consist of a period of aerobic exercise, followed by some form of stretching, and ending with a period of activity similar to the upcoming activity (36). A traditional warm-up program typically involves two key phases.

The first is a **general warm-up** period (48), which may consist of 5 minutes of low-intensity aerobic activity such as jogging, skipping, or cycling. The aim of this phase is to increase heart rate, increase and redistribute blood flow, increase deep muscle temperature, increase respiration rate, and decrease the viscosity of joint fluids (28). This phase historically has been followed by a period of general static stretching that aims to replicate the ranges of motion required for the upcoming activity, although this has been largely superseded by dynamic stretching protocols (48).

In the traditional approach, following the general warm-up is the **specific warm-up** period, which incorporates movements similar to the biomechanical movement patterns of the athlete's sport. This phase should also include rehearsal of the skills to be performed (97).

The whole warm-up should progress gradually and provide sufficient intensity to increase muscle and core temperatures without causing fatigue or reducing energy stores (62). Typically, it should last between 10 and 20 minutes, depending on the specific requirements of the context. For example, shorter time periods are far more common in the majority of training sessions, while longer time periods are more common in cases in which aspects of the warm-up are an integral part of the main session or when specific competition warm-ups are being carried out. The warm-up should ideally end no more than 15 minutes before the start of the subsequent activity; after this time the positive effects of the warm-up start to dissipate (30).

Targeted and Structured Warm-Ups

Activities undertaken within warm-ups using this structure of general and specific warm-ups vary considerably. Consequently, although athletes may be performing an appropriately structured warm-up, this cannot ensure that the warm-up itself adequately addresses the key variables required to optimize subsequent performance (36). Effective planning needs to carefully consider the specific requirements of the subsequent performance and how the warm-up can ensure that any necessary responses are elicited. Additionally, the needs and capacities of the athletes should be considered and be planned according to any specific logistical and environmental factors. It is also important that planning delineates between warm-ups used before competition and the more ubiquitous warm-ups used within a training session, because the ultimate objectives of both can be quite different. Whereas the key objective in a precompetition warm-up is to maximize performance in that competition, the training warm-up, as well as optimizing acute performance, can contribute to performance in other productive ways and across a wider range of timescales (49). Subsequently, the planning process for these is different. For example, planning for competition warm-up is totally focused on the short-term maximization of performance, whereas training warm-ups can also consider how the process contributes to the overall development of the athlete; this planning can therefore entail short-, medium-, and long-term considerations (48).

Medium- and long-term planning is a recent trend with regard to planning effective warm-ups (50). Athletes spend a great deal of training time warming up, so structures that allow optimal use of this time provide a potentially powerful tool for the coach (51). Effective warm-ups should be thought of as an integral part of a training session, not as a separate entity (51). With most training warm-ups lasting from 10 to 20 minutes, over a training cycle this contributes to a significant amount of training time; with effective planning, this

can contribute greatly to an athlete's overall development, as well as optimally preparing the athlete for the next session (48).

The warm-up is an integral part of the training session. Strength and conditioning professionals should plan warm-ups incorporating short-, medium-, and long-term considerations that will contribute to the overall development of the athlete.

In recent times, the use of structured and standardized warm-up systems such as the FIFA 11 has become more prevalent and emphasizes medium- and longer-term planning, often with the goal of injury reduction through the sequenced development of physical characteristics associated with a reduction in injury risk (2, 5, 33). The results of these have been encouraging and demonstrate the potential for longer term planning of warm-ups to contribute to athletic development (2, 5). However, despite the success of this approach, uptake has been surprisingly low; a potential reason for this has been the standardized structure in which activities are preprescribed that does not allow for flexible application (2).

Whereas structures such as the FIFA 11 are preprescribed, it is likely that a range of potentially effective warm-ups exist (89) and that these need to be able to adapt to specific situations and over time as the needs of the athletes change (89). Rather than use a single preprescribed approach, a more applicable approach can be to provide a structure around which to plan the warm-up, one that allows for variability and progression. The structure of general and specific warm-up, though valid, may need to be supplemented by an approach that allows for greater targeting of both short- and long-term performance. One such flexible structure that has been adopted by many coaches and that addresses all of the key aspects of an effective warm-up is the **raise, activate and mobilize, and potentiate (RAMP) protocol** (51). This builds on the general and specific structure and provides a suitable approach whereby

performance can be maximized in the short, medium, and long term (51). As the name indicates, raise, activate and mobilize, and potentiate are its three key phases.

The first phase of the RAMP protocol involves activities that raise the level of key physiological parameters while also focusing on enhancing the skill level of the athletes. While at first glance this phase could be considered analogous to the general warm-up, having the aim of elevating body temperature, heart rate, respiration rate, blood flow, and joint fluid viscosity via low-intensity activities, it is the latter focus on skill development that is very different. Here, unlike traditional general warm-up activities, the focus is not on generic aerobic exercises to elicit the required physiological responses but instead to use activities that simulate the movement patterns of the upcoming activity and develop the movement patterns or skill patterns the athlete will need to deploy within the sport. In this way, the session, from the outset, is specifically targeted at key movement patterns and skill capacities that contribute to longer term development, as well as providing for the required physiological effects of the activity (48). This also assists in the psychological preparation of the athlete for practice or competition because the session constantly addresses components that relate to performance.

The second phase, activating and mobilizing, could at first glance be confused with the static stretching component of a traditional warm-up. However, here again we see major differences in both the goal of the phase and subsequently the types of activities chosen. Here, a longer-term consideration is played out, whereby the focus is to develop an athlete's movement competency (48). By identifying key movement patterns required for athletic performance in both the subsequent session and in the athlete's overall development, such as squat patterns and lunge patterns, the phase can be built around the performance of these patterns. The focus here is not on **flexibility** (the range of motion around a joint) as a static component but on mobility (48). **Mobility** involves the capacity to

move freely and effectively through movements (67) and is a dynamic quality requiring a combination of motor control, stability, and flexibility; it more closely relates to the movement requirements an athlete will face in his or her sport and in the general activities of life (51). See [table 15.1](#) for a full example of a RAMP protocol.

Debate persists as to whether static stretching should be used within warm-up, and at present the evidence is equivocal. Some reviews of the literature on the effect of static stretching on performance question the practice (82, 83, 97), suggesting that it can compromise muscle performance (53). Studies have demonstrated negative effects of static stretching on performance in a range of parameters such as force production (9, 23-26, 27, 31, 75), power performance (21, 92, 96, 97), running speed (35), reaction or movement time (8), and strength endurance (70). It has been suggested that this may be associated with a reduction in the stiffness of the musculotendinous units caused by a change in intramuscular viscoelastic properties (5, 40) or to a reduction in efferent neural input to the muscles (40). However, other studies show no decrement in performance, and a more recent review by Kay and Blazevich (52) concludes that there is clear evidence that short-duration static stretching has no detrimental effect on subsequent performance unless stretches are held for greater than 60 seconds. Simic and coworkers (86) report that shorter-duration (<45 s) static stretching bouts result in less performance decrement but that the performance decrement still exists and may affect competitive performance. Therefore, in deciding whether to use static stretching within this phase of the warm-up, it is important that the strength and conditioning professional perform a benefit–risk analysis (59). An important factor to consider in designing this phase of the warm-up is the types of movements and ranges of motion required for the upcoming activity. Athletes in sports that require greater ranges of motion may need to spend more time in this phase of warm-up and use some elements of static stretching that may enhance technical elements of performance that depend

on range of motion (98), such as gymnasts and martial artists, than those athletes in sports and activities with a low range of motion requirement (42, 93). Additionally, if static stretching is used, research suggests that following it with a period of more intense activity may offset any potential decrements in performance (6).

However, as highlighted previously, effective warm-up planning should not be focused solely on short-term considerations but should also consider longer term outcomes, and this provides a different perspective to the debate. Because no consistent link has been shown between stretching and injury prevention (43, 44, 74, 85, 89), the focus in warm-ups should shift to performance. Exercises should be designed to contribute to preparation for the upcoming session but also enhance the athlete’s overall movement capacities. Exercises that encourage the athlete to develop and use key movement patterns are ideal activities within this phase of the warm-up (51).

The use of a range of exercises built around these key movement patterns provides all of these key advantages while also providing an opportunity to address any specific movement issues an athlete may demonstrate (51). Additionally, this approach helps maintain the temperature-related benefits of the raising phase of the warm-up (51). Furthermore, a number of joints can be integrated into a single activity, often including multiplanar movements similar to those that occur in sport. This integrated approach is also extremely time efficient, which can be important when training time is limited (49, 51). These advantages, together with the fact that these dynamic activities have been shown to improve subsequent running performance (34, 94), make these types of exercises the preferred activities during warming up for the majority of sports (51).

TABLE 15.1 Example Raise, Activate and Mobilize, and Potentiate Warm-Up Protocol

Phase	Action	Programming
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Phase	Action	Programming
Raise	Slow jog	3 × 10 yards
	Ankling (see page 675)	3 × 10 yards
Exercises performed moving from cone A to cone B and return	A-walk	2 × 10 yards
	Low amplitude dribble run	2 × 10 yards
	Side-shuffle (see page 659)	2 × 10 yards
	Moderate amplitude dribble run	2 × 10 yards
	A-skip (see page 676)	3 × 10 yards
Activation and mobilization	Y's and T's with band resistance	2 × 10 each
	Squat with overhead reach	10
	Lunge with overhead side reach (see page 400)	10
	Hip flexion to inverted hamstring curl	10
	Straight-leg march (see page 405)	6 each leg
	Spiderman crawl (see page 406)	2 × 10 yards
Potentiation	Squat jump to box	2 × 6
	Countermovement jump to box	2 × 6
	Low hurdle jump	2 × 5
	High hurdle jump	2 × 3
	Explosive dribble run	2 × 10 yards
	Light med ball slam	3 × 6
	First exercise progression	3 sets of 4 reps at 50%, 70%, and 90% of first set load

Phase	Action	Programming
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Note: For all drills that cover a distance, set up two cones (i.e., cone A and cone B) that are 10 yards (or meters) apart. Also, any distance measured in yards can be the same in meters; precision is not necessary.

What is crucial to note at this point is that while the first two phases have moved the body from a state of rest to a more prepared state, it is still unprepared for maximal performance, especially in activities that require high levels of force or speed application. This is why a warm-up must include a progressive sequence of exercises that increase the intensity of activity up to a level that will be experienced in the upcoming activity or sport; this is the potentiation phase. While this third phase, **potentiation**, could be considered as analogous to the specific warm-up, crucially its focus on the progressive intensity of activities is different. The quality of muscle contraction depends on its contractile history, and there exists a staircase or *treppe* effect, where consecutive contractions result in a net increase in contractile force (47, 60). This contractile history makes the motor neuron more sensitive to action potentials, allowing muscle to be activated with decreased neural input (47). This phase therefore deploys sport-specific or fitness capacity-specific (e.g., speed and agility exercises) activities that demonstrate a staged increase in intensity until the athlete is performing at the intensity required for the subsequent competition or training session. This phase is critical to subsequent performance, especially in activities that require high levels of speed, strength, and power, yet it is often, and surprisingly, omitted from traditional warm-ups. Indeed, the more speed or power necessary for the sport or activity, the more important the potentiation phase of warm-up (17), and including high-intensity dynamic exercises can facilitate subsequent performance (11, 14, 32, 96). Competition warm-up should include a progression of sport-specific activities that enable the athlete to maximize performance physiologically and psychologically (11). For the training warm-up, this can be targeted

to the upcoming session yet can also address the longer-term requirements of the athlete (48). With effective planning, this phase of the warm-up can be a key part of a training session, presenting an ideal opportunity to work on aspects of performance such as plyometrics, speed, and agility (51). Effective planning of this phase can allow considerable training time to be spent on key components of fitness without an overall increase in training duration (51). Indeed, it is this phase that ultimately determines the optimal length of the warm-up, because the potentiation phase can become a key part of the session, allowing the delivery of elements such as speed training and agility training; in these instances, the entire warm-up will be longer but will progress almost seamlessly to the main session.

Central Nervous System Preparatory Exercises

The potentiation phase of the warm-up is predicated on the history-dependent nature of muscle force production and involves a progressive series of exercises up to the intensity of the upcoming activity. This typically uses exercises similar to those in the main activity (e.g., a series of sprints before a football session). However, this history dependence also raises an important question: If a series of activities up to maximal intensity can elicit an enhanced neural output (e.g., enhanced motor unit recruitment or enhanced rate coding) and facilitate performance, can the use of augmented activities enhance this even further? This is the premise of a **central nervous system (CNS) preparatory exercise**, when a range of potential activities are used with the goal of enhancing preparedness for another activity via a positive transient performance response (45). Research suggests that an acute stimulus of low-volume exercise can acutely enhance neuromuscular performance, a concept known as **postactivation performance enhancement** (PAPE), and outlines the potential for residual force enhancement from prior activity (45).

The potential for residual force enhancement through CNS stimulation has been attributed to a range of physiological mechanisms, including the phosphorylation of myosin light chains, pennation angle changes in a muscle, recruitment of higher threshold motor units, and enhancement of the H-wave amplitude (91). It has also been suggested that PAPE could be attributed to an increase in the quantity of neurotransmitters released and their efficiency, resulting in enhanced muscle contractions together with an elevation in excitation potentials across synaptic junctions at the spinal cord (40, 91). However, no definitive answer emerges as to the mechanisms by which PAPE is elicited, and with so many potential contributory mechanisms, it is expected that there will be great variation in both the optimal stimulus and the probable response of an individual to any protocol. This potential individuality of response is made more probable by the fact that the preconditioning activity does not only have the potential to generate PAPE; it also has the potential to generate fatigue (the fitness fatigue theory). This balance between potentiation and fatigue must always be kept in mind when designing PAPE-based activities—indeed, throughout the design of a warm-up.

A range of modalities have been purported as having the potential to stimulate PAPE, including resistance training exercises, variable resistance exercise, isometric contractions, resisted sprinting, and assisted sprinting (72), with the majority of modalities reporting subsequent effects of sprint and/or jump performance (55). Most research into PAPE has focused on the use of heavy resistance exercise, predominantly at intensities of between 60% and 90% 1RM and using exercises such as the back squat. Similarly, the use of isometric muscle actions at a range of positions has been investigated. However, the challenge of this modality is its application given the necessity for equipment and the associated time to get athletes through the protocols. As a result, a more recent focus has been on the use of activities that are potentially more applicable such as resisted sprinting and plyometrics.

Across all protocols, the evidence to support the use of PAPE exercises is equivocal, with great variation on most modalities. Here the interaction of potentiation and fatigue will always be a challenge; for example, the activities that may generate the greatest potentiation are also those that are likely to result in the greatest fatigue. Definitive conclusions are made even harder by the variation in the warm-up used in the studies and whether the positive results are as much a result of an effective potentiation phase rather than evidence of PAPE. To further challenge the application of the phenomenon, it appears that the effects are also highly individual and relate to the interaction of a range of factors, such as the type of stimulus, the intensity of the stimulus, the volume of the stimulus, the time duration between stimulation and application, and the individual characteristics of the athlete. As a result, definitive guidelines on the application of PAPE in any warm-up (see the [sidebar](#)) should be considered tentative at best.

Much of the current research has focused on the acute application of CNS preparation. However, some research suggests that the positive effects of a bout of high-intensity exercise may last significantly longer than the windows that have been explored to date—perhaps up to 48 hours. As a result, the concept of precompetition priming has been proposed, using a low-volume, high-intensity session some hours or even days before a competition to prime the CNS, allowing for the exploitation of a potential residual force enhancement (45). However, the research on this is limited and does not fall under the purview of this chapter, where the focus will be on the use of CNS preparation in a warm-up.

While the use of PAPE in warm-ups may be alluring, the logistical challenges are many and often preclude effective application (55). Given the highly individual nature of PAPE, finding a single effective modality for a team of athletes will be extremely difficult, and as a result this application may be more suited to individual athletes rather than teams. Even here, logistical and operational considerations have to be taken into account. Equipment availability

at venues, time restrictions of when athletes need to be at certain positions, and so on limit the potential to apply modalities that require equipment, such as resistance training modalities, and it may be more prudent to look at methods that are more user friendly and can be incorporated into current warm-up modalities, such as the application of plyometrics. Here consideration needs to be placed not just on what modality generates the greatest potential enhancement but also on what can be effectively applied into warm-up routines. Throughout this analysis the potential for residual fatigue generation must be considered, which may outweigh the positive impact over an extended time period.

In terms of using PAPE within a training session warm-up, the interindividual differences will always present a challenge. What is important is that the case for PAPE does not blind the strength and conditioning professional to the overall goals of a session. For example, if a heavy squat stimulus was used to prime a jump performance but a rest period of 12 minutes was required to optimize the effect, the net effect on training volume within the session may far outweigh the potentially positive effects of PAPE. What is more likely is that the use of contrast-type methodologies in activities that may generate less PAPE but also require less recovery time between the preconditioning activity and the main activity may be more applicable. At this time, PAPE is an alluring concept but one still being studied for effective application. See [tables 15.2](#) and [15.3](#) for examples of CNS preparatory exercise combinations.

TABLE 15.2 Lower Body CNS Preparatory Exercise Combinations

Preconditioning activity	Variables	Main activity
Back squat (see page 439)	Load: 60%-90% 1RM Sets: 1-3 Reps: 1-3	Countermovement jump

Note: These are examples; the precise pre- and postconditioning activity should be adjusted to suit the needs of the session. Similarly, the precise variables should be adjusted to the needs of the athlete and the activity.

Preconditioning activity	Variables	Main activity
	Rest before activity: 3-12 min	
Resisted sprint (see page 684)	Load: 10%-60% body weight Distance: 10-25 m (11-27 yd) Reps: 1-3 Rest before activity: 2-5 min	Sprint
Depth jump (see page 615)	Load: Body weight Height: 30-90 cm (12-35 in.) Sets: 1-3 Reps: 1-3 Rest before activity: 1-5 min	Countermovement jump

Note: These are examples; the precise pre- and postconditioning activity should be adjusted to suit the needs of the session. Similarly, the precise variables should be adjusted to the needs of the athlete and the activity.

TABLE 15.3 Upper Body CNS Preparatory Exercise Combinations

Preconditioning activity	Variables	Main activity
Bench press (see page 430)	Load: 60%-90% 1RM Sets: 1-3 Reps: 1-3 Rest before activity: 3-12 min	Medicine ball throw
Push press (see page 463)	Load: 40%-80% 1RM Sets: 1-3 Reps: 1-3 Rest before activity: 3-12 min	Shot put

Note: These are examples; the precise pre- and postconditioning activity should be adjusted to suit the needs of the session. Similarly, the precise variables should be adjusted to the needs of the athlete and the activity.

Guidelines and Precautions for CNS Preparatory Exercises

- The stimulatory effect is likely greater with well-trained athletes with high speed and power capacities.
- Resistance-based protocols likely have the greatest potential for gain but also the greatest potential for fatigue.
- Plyometric and resisted sprint-based exercises may have lower potential but be more accessible for use in warm-ups.
- Higher intensities will likely generate greater potentiation but also greater fatigue and thus require greater recovery times.
- The closer the similarity between the preconditioning activity and the main activity, the higher the likely transfer to enhanced performance.
- The effects of all protocols will be highly variable and should be prescribed on an individual basis.
- Some athletes may not benefit from CNS stimulation.
- The logistical challenges of implementing protocols may outweigh any potential benefits.
- Any decision should be based on a wide and comprehensive cost-benefit analysis.

Flexibility

Flexibility is defined as the **range of motion (ROM)** that occurs about a joint. Flexibility can be expressed statically and dynamically. **Static flexibility** refers to the range of possible movement about a joint (28) and its surrounding muscles during a passive movement (40). Static flexibility requires no voluntary muscular activity; an external force such as gravity, a partner, or a machine provides the force for a stretch. **Dynamic flexibility**, on the other hand, refers to the available ROM during active movements and therefore requires voluntary muscular actions. Dynamic ROM is generally greater than static ROM. However, the relationship between static and dynamic ROM has historically been questioned and still remains largely unresolved, especially in relation to the fact that a normal range of motion cannot guarantee normal movement (19), indicating that ROM alone cannot determine the quality of any movement. Therefore, the direct transfer between measures of static flexibility and sport performance cannot be determined.

In determining the role flexibility can play in performance, it is important to look at the nature of the sport or activity being undertaken. Given that performance in the majority of sports is dynamic in nature, it is advised to look beyond the role of flexibility as a static quality and instead consider its contribution as a dynamic quality (66). Here, the concept of mobility introduced earlier can be a more enlightening term than *flexibility*, naturally reflecting this dynamic quality. Effective mobility requires the integration of aspects such as balance coordination, postural control coordination, and perception (19). In this way, mobility automatically frames flexibility as a dynamic quality, with the athlete required to demonstrate control, coordination, and force through any demonstrated ROM, essentially being analogous to dynamic flexibility. This is advantageous when considering the role of flexibility on performance. Indeed, enhanced ROM without appropriate motor control will not maximize performance (80).

Flexibility and Performance

Given the lack of a direct link between flexibility and movement capability, optimizing flexibility in relation to the specific activity rather than simply maximizing flexibility should be the focus of training (68). Clearly, the direct effects of flexibility on performance therefore depend on the types of activity undertaken, and it is always important to remember that the most flexible athlete is not always the most successful (84). Given that different sports and activities have specific requirements for ROM, it is likely that optimal levels of flexibility exist for each activity, with these related to the specific movements required within the particular sport or activity, together with the reduction of any potential injury risk that may present if an athlete lacks a specific ROM (93). As well as a consideration of the ROM itself, it is also important that the strength and conditioning professional keep in mind the kinetic and kinematic patterns required within the ROM, because this will also dictate the types of development methods used. The development of greater ROM should be linked with the requirements for enhancing performance, and this often involves the need to apply force over the required ROM and the ability to assume the key technical positions required for the sport. In some instances, greater ROM can be seen as the missing link in the athlete's power program, because the ability to apply force over a greater ROM can increase impulse by extending the time over which the athlete can apply force, which in turn contributes to enhanced performance in a range of sports (61). Thus, the strength and conditioning professional should look at the optimal ROM required of the athlete but also at the underlying goal of the actions and force patterns required through this ROM, and where greater ROM is required, this should be in tandem with the required force capacities to ensure that the athlete is optimally prepared for performance.

There are clearly optimal ranges of flexibility for different sports and activities, and injury risk may be increased when an athlete is unable to attain this range. Yet increased ROM is not a panacea, and

it is important to note that both inflexibility and hyperflexibility can increase injury risk (77, 89). For example, in contact sports, an excessive ROM at the shoulder or a laxity in the ligaments of the knee may predispose the athlete to injury in contact situations. Additionally, general flexibility imbalances between joints, which could increase injury risk, must be considered (87, 88).

Factors Affecting Flexibility

A number of anatomical and training-related factors affect flexibility. Training cannot alter some factors, such as joint structure, age, and sex. However, others can be manipulated and thus are important considerations in the design of flexibility development programs. A long-term program of stretching can have cumulative positive effects ranging from enhanced flexibility to enhanced strength (80). Strength and conditioning professionals should consider each athlete's unique combination of factors and sport requirements when recommending flexibility development exercises and programs.

Joint Structure

The structure of a joint determines its ROM (64). **Ball-and-socket joints**, such as the hip and shoulder, move in all anatomical planes and have the greatest ROM of all joints (64). The wrist is an **ellipsoidal joint** (an oval-shaped condyle that fits into an elliptical cavity), primarily allowing movement in the sagittal and frontal planes; its ROM is significantly less than that of the shoulder or hip (64). In contrast, the knee is a modified **hinge joint**, with movement primarily in the sagittal plane; its ROM is less than that of either a ball-and-socket joint or the ellipsoidal wrist joint. The type of joint, the shapes of the joint's articulating surfaces, and the soft tissues surrounding the joint all affect its ROM.

Age and Sex

Young people tend to be more flexible than older people (76), and females tend to be more flexible than males (76). Differences in

flexibility between young men and women may be due in part to structural and anatomical differences and the type and extent of activities performed. Older people undergo a process called **fibrosis**, in which fibrous connective tissue replaces degenerating muscle fibers (57). This is likely due to inactivity and a tendency to use less of the available ROM during movement. Just as older people can improve strength, they can improve flexibility with appropriate exercise.

Muscle and Connective Tissue

A range of body tissues can influence flexibility. For example, factors such as muscle tissue, the **musculotendinous unit (MTU)**, tendons, ligaments, fascial sheaths, joint capsules, and skin may limit ROM (28). The relative contributions of these to increased flexibility, both acutely and chronically, remain unclear and reflect the relative impacts of **elasticity** and **plasticity** (93). *Elasticity* refers to the ability to return to original resting length after a passive stretch, while *plasticity* refers to the tendency to assume a new and greater length after a passive stretch. Stretching exercises can positively affect connective tissues by taking advantage of their plastic potential, although the relative importance of each tissue type and the differences between individual responses remains unclear (12).

Stretch Tolerance

An important factor in determining an athlete's flexibility is stretch tolerance, or the ability to tolerate the discomfort of stretching (84). Individuals with a greater ROM tend to demonstrate a greater level of stretch tolerance and are thus able to tolerate a greater stretch load (12). One important feature of a regular stretching program is that it can increase an athlete's stretch tolerance, thus potentially permitting further increases in flexibility (7).

Neural Control

The control of an athlete's ROM is as much determined by central and peripheral nervous system inputs as by structural elements (66); see [chapter 1](#) for more detail. This system includes both afferent and efferent mechanisms, resulting in both reflexive and conscious activities that ultimately control the ROM an athlete is able to attain (see the *Proprioceptor* sidebars on [pages 376](#) and [378](#)). A key aspect of an effective flexibility program is the capacity to positively affect this system, allowing for elicitation of a greater ROM.

Resistance Training

A comprehensive and proper resistance training program may increase flexibility (69) and also assist in the development of force capacity through the enhanced ROM. Effective strength training protocols should be seen as an important adjunct to a stretching program (78). However, this comes with two provisos. The first proviso is that heavy resistance training with limited ROM during the exercises may decrease ROM (28). To prevent loss of ROM, an athlete should perform exercises that develop both agonist and antagonist muscles (16) and should exercise through the full available ROM of the involved joints.

Muscle Bulk

The second proviso is that a significant increase in muscle bulk may adversely affect ROM by impeding joint movement. An athlete with large biceps and deltoids, for example, may experience difficulty in stretching the triceps (28), causing difficulty in performing exercises such as racking a power clean or in performing sport-specific skills such as throwing at high velocities. Although altering the training program can decrease the amount of muscle bulk, this may not be advisable for large power athletes such as shot putters or American football offensive linemen. Strength and conditioning professionals should always keep the requirements of the athlete's sport in mind: In some activities, the need for muscle hypertrophy may supersede

the need for extreme joint mobility, but similarly, where ROM is crucial, the potential negative effects of muscle bulk should be considered and training programs planned accordingly.

Activity Level

An active person tends to be more flexible than an inactive one, although it is important to note that it is not activity level alone that improves flexibility but also the type of activity that influences flexibility (76). Stretching exercises, or exercises requiring the body to move through a full ROM, are essential if joint flexibility is to be maintained or increased. Similarly, resistance training, especially using full ranges of motion, can be an effective stimulus to maintain or increase ROM, and both men and women have successfully increased their flexibility as an outcome of properly designed resistance training programs (69).

Proprioceptor: Muscle Spindles

The **muscle spindles** are located within intrafusal muscle fibers that run parallel to extrafusal muscle fibers; they monitor changes in muscle length (40). During a rapid stretching movement, a sensory neuron from the muscle spindle innervates a motor neuron in the spine. The motor neuron then causes a muscle action of the previously stretched extrafusal muscle fibers; this is the **stretch reflex**. Stimulation of the muscle spindle and the subsequent activation of the stretch reflex should be avoided during stretching, because motion will be limited by the reflexive muscle action. If the muscle spindles are not stimulated, the muscle relaxes and allows greater stretch. Because of the very slow movement during static stretching, the stretch reflex is not invoked. Rapid (ballistic and dynamic) stretching movements may stimulate the muscle spindles, causing a stretch reflex.